INCH-POUND
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DETAIL SPECIFICATION

# INITIATOR, ARM FIRE, JAU-54A/A

*This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Department and Agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification covers the requirements for the manufacture and acceptance of the JAU-54A/A Arm Fire Initiator (3W84) (see 6.1).

## 2. APPLICABLE DOCUMENTS

**2.1 General.** The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in sections 3 and 4 of this specification, whether or not they are listed.

**2.2 Government documents.**

**2.2.1 Specifications, standards, and handbooks.** The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

| | |
|---|---|
| MIL-L-46225 | Lead Azide RD-1333 (For Use in Ammunition) |
| MIL-C-83124 | Cartridge Actuated Devices/Propellant Actuated Devices, General Design Specification for |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Engineering Documentation Branch (Code 4230), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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STANDARDS

DEPARTMENT OF DEFENSE

| | |
|---|---|
| MIL-STD-1168 | Ammunition Lot Numbering and Ammunition Data Card |
| MIL-STD-810B | Environmental Engineering Considerations and Laboratory Tests |
| MIL-STD-331 | Fuze and Fuze Components, Environmental and Performance Tests for |

(Unless otherwise indicated, copies of Federal and Military specifications and standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

**2.2.2 Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

SPECIFICATIONS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

| | |
|---|---|
| WS 5003 Rev F | HNS EXPLOSIVE |

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

| | |
|---|---|
| DL6106319 | Initiator, Arm Fire, JAU-54A/A |

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

| | |
|---|---|
| 841AS450 | Test Tip, Shielded Mild Detonating Cord |
| 857AS800 | Test Fixture Assembly |

(Application for copies should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Engineering Documentation Branch (Code 4230), 101 Strauss Avenue, Indian Head, MD 20640-5035.)

**2.3 Non-Government publications and drawings.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARD INSTITUTE/AMERICAN SOCIETY FOR QUALITY

| | |
|---|---|
| ANSI/ASQ Z1.4 | Sampling Procedures and Tables for Inspection by Attributes |

(Application for copies should be addressed to the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)



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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| ASTM E 1742 | Standard Practice for Radiographic Examination |
| ASTM E 545 | Standard Method for Determining Image Quality in Direct Thermal Neutron Radiographic Examination |

(Application for copies of ASTMs should be sent to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

**2.4 Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

### 3.1 Test procedures.

**3.1.1 First article tests.** When specified in the contract or purchase order (see 6.2), a sample shall be subjected to a first article inspection in accordance with 4.3.

**3.1.2 Lot acceptance tests.** Lot acceptance tests shall be performed in accordance with 4.4 and 4.5.

**3.2 Conformance to documents.** The JAU-54A/A Initiator and all its components shall be manufactured in accordance with DL6106319 and all documents listed thereon.

**3.3 Materials.** All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal. Only primary components (see 3.4) from a single lot shall be used in a production lot of initiators. One primary component production lot may be used in more than one initiator production lot.

**3.4 Primary components.** For the purposes of this specification, the primers, lead azide, and the hexanitrostilbene are considered primary components (see 6.4.1).

**3.4.1 Percussion primers.** Only percussion primers manufactured within the last 24 months or recertified within the last 12 months shall be used in these units.

**3.4.2 Lead azide.** Only lead azide conforming to MIL-L-46225, RD-1333, manufactured within the last 24 months or recertified within the last 12 months shall be used.

**3.4.3 Hexanitrostilbene (HNS II).** Only HNS II, Grade A, manufactured in accordance with WS 5003 Rev F shall be used in the transfer charge. The purity level of the HNS II batch being used shall be certified by the manufacturer. A copy of the certification shall be furnished with the contract data.

**3.5 Transfer charge.** All transfer charges shall be manufactured with primary components which meet the requirements of 3.4. Individual transfer charges shall be permanently marked with a unique serial number.



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**3.5.1 Visual examination.** All transfer charges shall be subjected to visual examination. Any transfer charge exhibiting cracks, dents, damaged closure disk, defective or incomplete weldments, missing serial number or any other defect which could adversely affect form, fit, or function shall be rejected. Examination shall be performed in accordance with 4.6.3.1.

**3.5.2 Radiographic examination.** All transfer charges shall be x-rayed and n-rayed and shall be free of the following defects when examined in accordance with 4.6.4: missing, defective or improperly assembled components; foreign matter; cracks; abnormal voids or other abnormalities which could adversely affect the form, fit or function of the transfer charge.

**3.5.3 Leak test.** All transfer charges shall pass a dry gas leak test. Transfer charges which exhibit a leak rate greater than 1.0 E-5 standard cc/sec of air at a pressure differential of 1.0 ± 0.1 atmosphere shall be rejected. Documentation certifying that all transfer charges successfully passed this leak test shall be supplied by the manufacturer. Testing shall be performed in accordance with 4.6.1.

**3.6 Initiator assembly.**

**3.6.1 Transfer charge.** Only transfer charges meeting the requirements of the applicable drawings and manufactured in accordance with 3.5 shall be used in initiators. A marriage log shall be kept recording the serial number of the transfer charge installed in each initiator assembly. A copy of the marriage log shall be furnished with the contract data.

**3.6.2 Visual examination.** All initiators shall be visually inspected for cracks, splits, missing safety pin, incomplete or unclear markings or other defects which might prevent proper mating with connecting parts or proper functioning of the device. Any initiator exhibiting one or more of these defects shall be rejected. Examination shall be in accordance with 4.6.3.2.

**3.6.3 Radiographic examination.** All initiators shall be x-rayed in accordance with 4.6.4. Any initiator exhibiting missing, defective or improperly assembled components, foreign matter, cracks, abnormal voids or other abnormalities which could adversely affect the form, fit or function shall be rejected. Initiators shall be positioned so that the axis of the quick release pin is perpendicular to the x-ray beam. X-rays shall be identified by date, contract number, part number, lot number and serial number.

**3.6.4 Residual magnetism.** All initiators shall be subjected to the residual magnetism test as specified in 4.6.2. Any initiator that deflects the compass point more than 5 degrees in any direction shall be degaussed and retested until it passes or it shall be rejected.

**3.7 Environmental conditions.** All initiators shall be capable of meeting the ballistic requirements of 3.8 after exposure to the environmental conditions of 3.7.2 through 3.7.7 when a first article or modified lot acceptance conformance inspection is required. Test units subjected to the forty foot drop test are not ballistically tested. Only initiators meeting the requirements of 3.6 shall be subjected to environmental conditions.

**3.7.1 Forty foot drop test.** To ensure safety of handling, the initiator shall not fire when dropped from a height of 40 feet. Initiators subjected to the forty foot drop test shall not be ballistically tested. Testing shall be in accordance with 4.3.2.2.1.

**3.7.2 Six foot drop test.** Initiators shall not fire when dropped from a height of 6 feet in accordance with 4.3.2.2.2. After being subjected to the 6 foot drop test the initiators shall meet the requirements of 3.8 when ballistically tested as specified in Table I.



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**3.7.3 Temperature/Humidity cycling test.** The test initiators shall be subjected to a 28 day temperature/humidity (T/H) test in accordance with 4.3.2.2.3. After being subjected to the T/H environment the initiators shall meet the requirements of 3.8 when ballistically tested as specified in Table I.

**3.7.4 Shock test.** The test initiators shall not fire or suffer damage which could adversely affect the form, fit or function of the initiator when subjected to the shock environments in accordance with 4.3.2.2.4. After being subjected to the shock environment the initiators shall meet the requirements of 3.8 when ballistically tested as specified in Table I.

**3.7.5 Vibration test.** The test initiators shall not fire or suffer damage which could adversely affect the form, fit or function of the initiator when subjected to the vibration environment in accordance with 4.3.2.2.5. After being subjected to the vibration environment the initiators shall meet the requirements of 3.8 when ballistically tested as specified in Table I.

**3.7.6 Dust test.** The test initiators shall be exposed to a dust environment as specified in 4.3.2.2.6. After being subjected to the dust environment the initiators shall meet the requirements of 3.8 when ballistically tested as specified in Table I.

**3.7.7 Salt Fog test.** The test initiators shall be capable of withstanding exposure to salt fog as specified in 4.3.2.2.7 and shall meet the requirements of 3.8 when ballistically tested as specified in Table I.

**3.8 Ballistic requirements.** Initiators from each lot, selected in accordance with 4.5.1, shall meet the requirements of 3.8.1 through 3.8.4 when tested in accordance with 4.6.6.

**3.8.1 Arming torque.** The torque required to rotate the Shaft Assembly 90 degrees from the safe position to the armed position with the quick release pin (safety pin) removed shall be 6 to 12 inch-pounds. Failure to meet this requirement shall result in rejection of the lot represented. Testing shall be performed in accordance with 4.6.6.2.

**3.8.2 Actuation force.** The force required to pull the Shaft Assembly from the initiator body when the Shaft Assembly is in the armed position shall be 20 to 35 pounds. Failure to meet this requirement shall result in rejection of the lot represented. Testing shall be performed in accordance with 4.6.6.3.

**3.8.3 Ballistic output.** The output of each initiator tested shall be verified by confirming that the dent produced in the witness plate is 0.001 inch deep minimum. Testing shall be performed in accordance with 4.6.6.4. Failure to meet this requirement shall result in rejection of the lot represented.

**3.8.4 Mechanical failure.** The initiator shall function in accordance with the performance requirements specified herein without mechanical failure (see 6.4.2). Failure to meet this requirement shall result in rejection of the lot.

**3.9 Workmanship.** Initiators shall be constructed and finished in a manner to assure compliance with all the provisions of this specification and the applicable drawings. Special attention shall be directed to dimensions, finishes, coatings and cleanliness.



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## 4. VERIFICATION

### 4.1 Classification of inspections.

a. First article inspection (see 4.3)
b. Production conformance inspections (see 4.4)
c. Lot acceptance conformance inspections (see 4.6)

**4.2 Inspection conditions.** Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.6.

**4.3 First article inspections and tests.** First article inspection shall be performed on units which have been manufactured in the same manner and using the same materials, equipment, processes, and procedures as to be used in the regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply. The first article shall be selected in accordance with 4.3.1 and inspected in accordance with 4.3.2. Any production prior to written notification of acceptability of the first article sample by the contracting agency shall be at the contractor's risk.

**4.3.1 Sample size.** Unless otherwise specified (see 6.2) the first article sample shall consist of 34 assembled initiators and 10 unassembled sets of components for dimensional and visual inspection and testing.

**4.3.2 Inspection routine.**

**4.3.2.1 Component inspection.** Ten (10) sets of initiator components shall be inspected to verify conformance with the requirements of 3.3. Failure to meet these requirements shall result in rejection of the first article sample.

**4.3.2.2 Initiators.** Thirty-six (32) initiators shall be tested as described in Table I. The 32 initiators shall be randomly divided into 7 groups, A through G. The quantity of initiators in each group shall be as specified in Table I. Two (2) additional initiators shall be retained for investigative purposes. All radiographic requirements listed in Table I except the post-environmental initiator x-ray are the responsibility of the contractor. The radiographs shall be delivered to the procuring activity with the first article. A first article sample acceptable for functional and environmental testing is defined as a sample that has met the requirements of 3.2 through 3.5 of this specification. Unless otherwise specified the environmental tests specified in Table I shall be performed in accordance with MIL-C-83124. The ballistic tests shall be performed using the test fixture specified in 4.6.6.1. Acceptance shall be based on integrity of the parts and legibility of markings after environmental exposures and acceptable ballistic performance in accordance with the requirements of 3.8.

**4.3.2.2.1 Forty foot drop test.** Test initiators shall be dropped from a height of 40 to 41 feet in accordance with test 103.2 of MIL-STD-331. The impact surface shall be a 2 inch minimum thick steel plate embedded in concrete. The shipping caps/plugs and the quick release safety pin shall be removed and the handle shall be in the unarmed position. The drop orientation shall be equally divided between handle-up and handle-down. If any initiator fails to meet the requirements of 3.7.1 the first article sample shall be rejected.

**4.3.2.2.2 Six foot drop test.** Test initiators shall be dropped from a height of 6 feet + 2 inches – 0 inches onto a 2 inch minimum thick steel plate embedded in concrete with the shipping caps/plugs and the quick



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release safety pin removed. The handle shall be in the unarmed position. The drop orientation shall be equally divided between handle-up, handle-down and horizontal orientation. If any initiator fails to meet the requirements of 3.7.2 the first article sample shall be rejected.

**4.3.2.2.3 Temperature/Humidity cycling test.** Test initiators shall be subjected to the temperature/humidity environments described in MIL-C-83124, paragraph 4.5.11.7 except the altitude shall be zero feet (14.7 psia). If any initiator fails to meet the requirements of 3.7.3 the first article sample shall be rejected.

**4.3.2.2.4 Shock test.** Test initiators shall be subjected to shock tests in accordance with method 516, procedure I of MIL-STD-810B with the safety pin removed. The shock pulse wave shall be a terminal peak sawtooth. The peak amplitude shall be 20g and the duration shall be 11 milliseconds. If any initiator fails to meet the requirements of 3.7.4 the first article sample shall be rejected.

**4.3.2.2.5 Vibration test.** Test initiators shall be subjected to the vibration test in accordance with method 514, procedure I, parts 1, 2, and 3 using curve Z up to 2000 cycles per second of MIL-STD-810B. The tee-handle shall be in the safe position but the safety pin shall be removed. Test units shall be vibrated at the temperatures specified in table I. If any initiator fails to meet the requirements of 3.7.5 the first article sample shall be rejected.

**4.3.2.2.6 Dust test.** Test initiators shall be subjected to the dust test in accordance with method 510 of MIL-STD-810B. If any initiator fails to meet the requirements of 3.7.6 the first article sample shall be rejected.

**4.3.2.2.7 Salt fog test.** Test initiators shall be subjected to a salt fog in accordance with method 509 of MIL-STD-810B. If any initiator fails to meet the requirements of 3.7.7 the first article sample shall be rejected.

**4.3.2.3 Failure.** Failure of any initiator or transfer charge to meet the requirements of DL6106319 or this specification shall be cause for rejection of the first article. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any or all of the stated requirements. In the event of failure of any sample, the Government reserves the right to require the contractor to take corrective action and submit a new sample or portion thereof. Until a first article inspection sample is accepted, the contractor is not authorized by the Government to initiate regular production unless otherwise directed in writing by the procuring activity.



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TABLE I. First Article Tests

| Test/Inspection | Requirement paragraph | Test paragraph | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|---|---|
| Quantity | | | 4 | 3 | 9 | 4 | 4 | 4 | 4 |
| Transfer Charge Radiographic Examination | 3.5.2 | 4.6.4 | 4 | 3 | 9 | 4 | 4 | 4 | 4 |
| Transfer Charge Leak Test | 3.5.3 | 4.6.1 | by cert. | by cert. | By cert. | by cert. | by cert. | by cert. | by cert. |
| Visual Inspection | 3.6.2 | 4.6.3.2 | 4 | 3 | 9 | 4 | 4 | 4 | 4 |
| X-ray | 3.6.3 | 4.6.4 | 4 | 3 | 9 | 4 | 4 | 4 | 4 |
| Magnetism | 3.6.4 | 4.6.2 | 4 | 3 | 9 | 4 | 4 | 4 | 4 |
| Forty foot drop | 3.7.1 | 4.3.2.2.1 | 4 | | | | | | |
| Six foot drop | 3.7.2 | 4.3.2.2.2 | | 3 | | | | | |
| Temp./Hum. | 3.7.3 | 4.3.2.2.3 | | | 9 | | | | |
| Shock | 3.7.4 | 4.3.2.2.4 | | | | 4 | | | |
| Vibration | 3.7.5 | 4.3.2.2.5 | | | | | 4 | | |
| Dust | 3.7.6 | 4.3.2.2.6 | | | | | | 4 | |
| Salt Fog | 3.7.7 | 4.3.2.2.7 | | | | | | | 4 |
| X-ray | 3.6.3 | 4.6.4 | | 3 | 9 | 4 | 4 | 4 | |
| Ballistic test (-65°F) | 3.8 | 4.6.6 | | | 3 | 1 | 1 | 1 | 1 |
| Ballistic test (70°F) | 3.8 | 4.6.6 | | 3 | 3 | 2 | 2 | 2 | 2 |
| Ballistic test (200°F) | 3.8 | 4.6.6 | | | 3 | 1 | 1 | 1 | 1 |
| Discard | | | 4 | | | | | | |

**4.4 Production conformance inspection.** Production conformance inspection for production lots shall consist of 100 percent inspection by the contractor of the characteristics classified on the applicable drawings and the tests and examinations of Table II.

TABLE II. Production Conformance Tests And Examinations.

| Item | Nature of test | Requirement | Test |
|---|---|---|---|
| Primary Component | Single Lot Verification | 3.4 | |
| Transfer Charge | Lot Acceptance Verification | 3.6.1 | |
| Primer | Age and Inspection | 3.4.1 | |
| Transfer Charge | Leakage | 3.5.3 | 4.6.1 |
| Initiator | Magnetism | 3.6.4 | 4.6.2 |
| Initiator | Visual Inspection | 3.6.2 | 4.6.3.2 |
| Initiator | X-ray | 3.6.3 | 4.6.4 |
| Initiator | Preservation-packaging, packing and marking | Section 5 | |

**4.4.1 Lot size.** The lot size shall be 1200 maximum. The lot shall include 2 units which will be retained for investigative purposes and the test sample units shall be selected as specified in 4.5.1. Initiators required for all test purposes are selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract.



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**4.5 Lot Acceptance Inspection.** Lot acceptance samples shall be subjected to the inspections and tests specified in Table III.

**4.5.1 Sampling.** The test sample size shall be as specified in Table IV. The test sample units shall be randomly selected in accordance with ANSI/ASQC Z1.4. Unless otherwise specified in the contract, all samples for test purposes shall be provided at the contractors' expense.

TABLE III. Lot Acceptance Tests

| Test/Inspection | Requirement paragraph | Test paragraph | A | B | C |
|---|---|---|---|---|---|
| Quantity | | | 1/3 Sample | 1/3 Sample | 1/3 Sample |
| Transfer Charge Radiographic Examination | 3.5.2 | 4.6.4 | Vendor Supplied | Vendor Supplied | Vendor Supplied |
| Transfer Charge Leak Test | 3.5.3 | 4.6.1 | by cert. | by cert. | by cert. |
| Visual Inspection | 3.6.2 | 4.6.3.2 | all | all | all |
| X-ray | 3.6.3 | 4.6.4 | all | all | all |
| Magnetism | 3.6.4 | 4.6.2 | all | all | all |
| Ballistic test (-65°F) | 3.8 | 4.6.6 | all | | |
| Ballistic test (70°F) | 3.8 | 4.6.6 | | all | |
| Ballistic test (200°F) | 3.8 | 4.6.6 | | | all |

TABLE IV. Production Lot Test Sample Size

| Lot size | 1-25 | 26-50 | 51-90 | 91-150 | 151-280 | 281-500 | 501-1200 |
|---|---|---|---|---|---|---|---|
| Sample size | 6 | 9 | 15 | 21 | 33 | 51 | 81 |

**4.6 Lot acceptance conformance inspections.**

**4.6.1 Leak test.** All transfer charges shall be tested with a dry gas leak detector of sufficient sensitivity to ascertain whether the cartridges meet the requirement of 3.5.3.

**4.6.2 Residual magnetism.** The assembly shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the test initiator six inches from the end and at the same height as the compass in a north-south horizontal position with the compass set in an east-west heading. The initiator shall be moved at a speed no greater than 10 feet per minute past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly past the compass. This procedure shall be repeated for 90 degree rotation of the initiator. Failure to meet the requirement of 3.6.4 will result in the rejection of the initiator.

**4.6.3 Visual inspection.**

**4.6.3.1 Transfer charge visual inspection.** Visually examine all transfer charges produced under contract and reject for any of the defects specified in 3.5.1. Failure to meet the requirement of 3.5.1 shall result in the rejection of the transfer charge.

**4.6.3.2 Initiator assembly visual inspection.** Visually examine all initiator assemblies produced under contract and reject for any of the defects specified in 3.6.2. Failure to meet the requirement of 3.6.2 shall result in the rejection of the initiator.



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**4.6.4 Radiographic examination.** Radiographically examine the specified items in accordance with ASTM E 1742 (x-ray) or ASTM E 545 (n-ray) as applicable. One (1) unit may be used as the penetrometer. Any observable imperfection in components or assembly shall be cause for rejection. All items shall be identified with serial numbers in consecutive numerical order beginning with 001 in each production lot prior to exposure. The items shall be arranged on boards or trays in consecutive order with any missing serial numbers identified on the radiographic plate. Each radiograph shall carry a permanent identification of the items displayed thereon. The radiographic identification shall include item nomenclature, the contract number, the complete lot number assigned to that lot, and the span of serial numbers displayed. Defective items found during radiographic review are to be marked on the radiograph and, prior to shipment, all defective items shall be removed from the production lot. Radiographs of the entire production lot and a description of the defects of removed items shall accompany the ballistic sample to the activity conducting the production lot acceptance tests. Failure to meet the requirement of 3.5.2 or 3.6.3, as applicable, shall result in the rejection of that item.

**4.6.4.1 Transfer charge radiographic examination.** Position each transfer charge such that its center plane passing through both primers is perpendicular to the radiographic beam. Radiographs shall be identified by date, contract number, part number, lot number and serial number.

**4.6.4.2 Initiator radiographic examination.** Position each initiator such that the centerline of the initiator and the centerline of the safety pin are both perpendicular to the radiographic beam.

**4.6.6 Ballistic tests.**

**4.6.6.1 Tests.** Sample initiators shall be randomly selected to be representative of the complete lot. One third of samples shall be temperature conditioned at each of the following temperatures for 4 hours minimum but not to exceed 24 hours maximum: –65 ± 5°F, 70 ± 5°F and 200 ± 5°F. The 24 hour maximum does not apply to the 70°F conditioning. Each sample shall be fired in the test fixture described in drawing 857AS800 within 3 minutes after removal from conditioning. If the 3 minute limit is exceeded the unit shall be returned to a 70 ± 5°F environment for 4 hours minimum and then reconditioned in the intended environment for the designated time. The initiators shall then be actuated and shall meet the requirements of 3.8.

**4.6.6.2 Arming torque.** Test initiators shall be mounted in the test fixture described in drawing 857AS800 with the quick release pin (safety pin) removed and instrumentation shall be provided to measure the arming torque required to rotate the Shaft Assembly 90 degrees from the safe position to the armed position. This torque requirement does not apply to the torque required to rotate the handle from the armed position back to the safe position. Failure to function and meet the requirements of 3.8.1 shall result in the rejection of the lot.

**4.6.6.3 Actuation force.** Test initiators shall be mounted in the test fixture described in drawing 857AS800 with the quick release pin (safety pin) removed and instrumentation shall be provided to measure the force required to pull the Shaft Assembly out of the initiator. Failure to function and meet the requirements of 3.8.2 shall result in the rejection of the lot.

**4.6.6.4 Ballistic output.** A 0.062 inch thick witness plate of aluminum alloy 6061-T6 shall be secured to the exposed mild detonating cord (MDC) of the output test tip, 841AS450-1. Failure to function and meet the requirements of 3.8.3 will result in the rejection of the lot.



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**4.6.6.5 Test failure.** Any failures or defects found during lot acceptance testing may cause the lot to be rejected. If test failure is not attributed to the initiator, the data may be disregarded for that firing and a retest permitted with a new initiator.

**4.7 Retest.** There shall be no retest, except as defined and allowed in 4.6.6.5. Any defects or failures during lot acceptance testing may be cause for rejection.

## 5. PACKAGING

**5.1 Packaging.** For acquisition purposes, the packaging requirements shall be specified in the contract or purchase order (see 6.2 and 6.5). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military's Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

**6.1 Intended use.** The JAU-54A/A is a cockpit mounted tee handle initiator used in the canopy removal system of AH-1 helicopters. The canopy removal system is only used in military aircraft, therefore the initiator has no commercial application. The JAU-54A/A is a one for one replacement for the JAU-54/A.

**6.2 Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2 and 2.3).

c. Inspection conditions if other than specified in 4.2.

d. First article sample size, if other than as specified (see 4.3.1)

e. Whether first article inspection is required and, if so, specify the test activity (see 3.1 and 4.3). A first article sample may be required at the contracting agency's discretion or if:
   1. two or more years have passed since the manufacturer last produced the JAU-54A/A,
   2. there has been a significant turnover in personnel,
   3. there have been major changes to the production facility or
   4. there has been transfer of production to a new facility.

f. Lot size, if other than as specified (see 4.4.1).

g. Packaging (see 5.1 and 6.5).

h. Sampling, if other than as specified (see 4.5.1, and 6.6)



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i. Date and place of delivery

j. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as a part of the description of work to be accomplished by a Government activity the safety precautions requirements of "Ammunition and Explosives," OP 5, are applicable.

**6.2.1 Additional Contract Data Requirements required by this specification.**

a. A copy of the certification of the quality of the HNS used in the transfer charge should be supplied (3.4.3).

b. Each transfer charge should have a unique serial number permanently marked on it (3.5).

c. A marriage log should be furnished identifying which transfer charge is installed in each initiator (3.6.1).

d. Documentation certifying that all transfer charges successfully passed the leak test should be supplied by the manufacturer (3.5.3).

**6.3 First article.** When a first article inspection is required, the items should be a first article sample. The first article should consist of 34 assembled initiators and 10 complete sets of components. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitation(s) for bid(s) should provide that the U.S. Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the U. S. Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior U.S. Government approval is presently appropriate for the pending contract.

**6.4 Definitions.** For the purpose of this specification, the following definitions apply.

**6.4.1 Primary component.** Components from a single lot.

**6.4.2 Mechanical failure.** Any deformation or breakage of a part or separation of parts which is other than a design function.

**6.4.3 Test fixtures.** The ballistic testing is to be conducted in a test fixture that conforms to Drawing 857AS800.

**6.5 Packaging.** The inner and intermediate packing and packaging should be in accordance with SPI (AM) P1377-13.

**6.6 Subject term (keyword) listing.**

Propellant
Canopy removal



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AH-1 helicopter

**6.7 Changes from previous issue.** Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity:
Navy - OS
(Project 1377-0161)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-DTL-32059A | 2. DOCUMENT DATE (YYMMDD)<br>25 June 2001 |
|---|---|---|

3. DOCUMENT TITLE
INITIATOR, ARM FIRE, JAU-54A/A

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (*Last, First, Middle Initial*) | b. ORGANIZATION | |
|---|---|---|
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| a. NAME<br>Commander, Indian Head Division<br>Naval Surface Warfare Center | b. TELEPHONE (*Include Area Code*)<br>(1) Commercial: 301-744-4700<br>(2) DSN: 354-4700 |
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